

Web Programming

Web Programming

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CSS first steps

What is CSS?

Lecture: What is CSS?

Module: CSS first steps

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Filter

CSS — Styling the web

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Learn to style HTML using CSS

Cascading Style Sheets — or [CSS](#) — is the first technology you should start learning after [HTML](#). While HTML is used to define the structure and semantics of your content, CSS is used to style it and lay it out. For example, you can use CSS to alter the font, color, size, and spacing of your content, split it into multiple columns, or add animations and other decorative features.

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Prerequisites

You should learn the basics of HTML before attempting any CSS. We recommend that you work through our [Introduction to HTML](#) module first.

Once you understand the fundamentals of HTML, CSS at the same time, moving back and forth between the two is more interesting and much more fun to learn when you apply CSS, and you can't learn CSS without

In this article

- Prerequisites
- Modules
- Solving common CSS problems
- "CSS is weird"
- See also



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<https://tinyurl.com/znuz8u2u>

Introduction

Objective of the lecture

Objective: This lecture explains what CSS is and covers some key terms about the language.

Introduction

- When a **web document** is **loaded** in the **viewport** of a **web browser**, what you **see** is the **browser's default styles** in action.
- **Default styles** are basic **styles** that the **browser applies** to **HTML** to make **sure** it will be basically **readable even** if **no explicit styling** is **specified** by the **author** of the **page**.

Browser defaults

- Using CSS you can **control** exactly **how HTML elements look in the browser, presenting your markup using whatever design you like.**

Browser defaults

The browser will style HTML documents using an internal stylesheet. This ensures that headings are larger than normal text, links are highlighted and structures such as lists and tables are understandable.

Paragraphs are spaced out. List items get a bullet or number, [Links are highlighted and underlined.](#)

- Item One
- Item Two

A level 2 heading

You can change all of this with CSS.

What is CSS for?

What is CSS for?

- **CSS** is a **language** for **specifying** how **documents** are **presented** to users.
- A **document** is usually a **text file structured** using a **markup language**.
 - ✓ **HTML** is the **most common markup language**, but you may also come across other such as **SVG** or **XML**.

Note

Note: A **browser** is sometimes called a **user agent**, which basically means a **computer program** that represents a **person** inside a **computer system**. **Browsers** are the **main type of user agent** we think of when talking about **CSS**, however, it is not the only one. There are **other user agents** available — **such** as those which **convert HTML** and **CSS documents** into **PDFs** to be **printed**.

CSS Syntax

CSS syntax

- CSS is a rule-based language.
 - ✓ You define rules specifying groups of styles that should be applied to particular elements or groups of elements on your web page.
- Example
 - ✓ "I want the main heading on my page to be shown as large red text."

```
1 h1 {  
2   color: red;  
3   font-size: 5em;  
4 }
```

CSS syntax

- The **rule opens with a selector**.
 - ✓ It **selects the HTML element** that we are **going to style**.
- **Declarations** take the **form of property and value** pairs.

```
1 h1 {  
2   color: red;  
3   font-size: 5em;  
4 }
```

CSS syntax

- CSS properties have **different** allowable **values**, depending on which **property** is being **specified**.
- A CSS **stylesheet** will contain **many rules**, written **one after the other**.

```
1  h1 {  
2      color: red;  
3      font-size: 5em;  
4  }  
5  
6  p {  
7      color: black;  
8  }
```

Note

Note: You can find **links** to **all the CSS property pages** (along with other CSS features) listed on the **MDN CSS reference**. Alternatively, you should get used to searching for "**mdn *css-feature-name***" in your favorite **search engine** whenever you need to find out more information about a **CSS feature**. For example, try searching for "**mdn color**" and "**mdn font-size**"!

Safari File Edit View History Bookmarks Develop Window Help

https://developer.mozilla.org/en-US/docs/Web/CSS/Reference

CSS reference - CSS: Cascading Style Sheets | MDN

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CSS reference

Web technology for developers ▸ CSS

On this Page

- Basic rule syntax
- Keyword index
- Selectors
- Concepts
- DOM-CSS / CSSOM
- See also

Basic rule syntax

Style rule syntax

```
style-rule ::=  
  selectors-list {  
    properties-list  
  }
```

Related Topics

CSS

CSS Reference

The individual property pages on MDN give you a quick way to look up properties and their values when you forget or want to know what else you can use as a value.

<https://mzl.la/2UA0eAn>

CSS modules and specification

CSS specifications

- All web standards technologies are defined in giant documents called specifications ("specs")
 - ✓ HTML, CSS, JavaScript, etc.
- Specs are published by standards organizations and define precisely how those technologies are supposed to behave.
 - ✓ Organizations such as the W3C, WHATWG, ECMA or Khronos.

CSS specifications

- CSS is no different
 - ✓ It is developed by a group within the W3C called the CSS Working Group.
- The CSS Working Group is made of representatives of browser vendors and other companies who have an interest in CSS.

CSS specifications

- The **CSS Working Group** develops and specifies new CSS features.
- A **key thing** about CSS is that **everyone works to never change things** in a way that would **break old websites**.
 - ✓ A **website built** in **2000**, using the **limited CSS** available then, should still be **usable** in a **browser today**!

CSS specifications

- It is **likely** that you will **find** the **CSS specs overwhelming**.
 - ✓ They are **intended** for **engineers** to **use** in order to **implement support** for the **features** in **user agents**.
 - ✓ **Not** for **web developers** to **read** to **understand CSS**.
- Many **experienced developers** would much rather refer to **MDN documentation** or **other tutorials**.

The CSS specification is broken down into modules (separate documents about some part of the specification).

CSS Backgrounds and Borders

Web technology for developers › CSS: Cascading Style Sheets › CSS Backgrounds and Borders

English ▼

On this Page

[Reference](#)

[Tools](#)

[Specifications](#)

CSS Backgrounds and Borders is a module of CSS that lets you style elements' backgrounds and borders. Backgrounds can be filled with a color or image, clipped or resized, and otherwise modified. Borders can be decorated with lines or images, and made square or rounded. Additionally, element boxes can be decorated with a shadow.

Related Topics

[CSS](#)

[CSS Reference](#)

[CSS Backgrounds and Borders](#)

▼ [Guides](#)

[Scaling background images](#)

[Using multiple backgrounds](#)

Reference

CSS Properties

[background](#)

[background-attachment](#)

[background-clip](#)

[background-color](#)

[border-bottom-right-radius](#)

[border-radius](#)

[border-bottom-style](#)

[border-bottom-width](#)

[border-collapse](#)

<https://mzl.la/2UA0eAn>

You can select and see the results (on the right) of different examples of the background property. There are examples like these for all CSS properties.

backg

Web technology fo

On this Page

Constituent prop

Syntax

Accessibility concerns

Formal definition

Formal syntax

Examples

Specifications

Browser compatibility

See also

Related Topics

CSS

CSS Reference

CSS Backgrounds and Borders

▼ Guides

Resizing background images with
background-size

CSS Demo: background

Reset

```
background: green;
```

```
background: content-box radial-gradient(crimson, skyblue,
```

```
background: no-repeat url("../media/examples/lizard.pr
```

```
background: left 5% / 15% 60% repeat-x url("../media/e
```

```
background: center / contain no-repeat url("../med  
#eee 35% url("../media/examples/lizard
```



<https://mzl.la/2UA0eAn>

Browser support

Browser support

- It is **unusual** for **all browsers** to **implement** a **feature** at the **same time**.
 - ✓ **There is usually** a **gap** where you can use **some part of CSS** in **some browsers** and **not** in **others**.
- For this **reason**, being able to **check implementation status** is useful.

Browser support

- On each property page on MDN you can see the status of the property you are interested to see if you will be able to use it on a website.
- What follows is the compatibility data chart for the CSS `font-family` property.

Update compatibility data on GitHub

	Desktop						Mobile					
	Chrome	Edge	Firefox	Internet Explorer	Opera	Safari	Android webview	Chrome for Android	Firefox for Android	Opera for Android	Safari on iOS	Samsung Internet
<code>font-family</code>	1	12	1★	3	3.5	1	1	18	4	10.1	1	1.0
<code>system-ui</code>	56	79	No	No	43	9★	56	56	No	43	9	6.0

What are we missing?

Full support

No support

★ See implementation notes.

⚡ Uses a non-standard name.

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Can I use... Support tables for HTML5, CSS3, etc

CSS property: font-family

Usage % of all users Global 95.57%

Current aligned Usage relative Date relative Apply filters Show all ?

IE	Edge	Firefox	Chrome	Safari	Opera	iOS Safari	Opera Mini	Android Browser	Opera Mobile	Chrome for Android	Firefox for Android	UC Browser for Android	Samsung Internet	QQ Browser	Baid Brow:
6-10	12-79	2-73	4-79	3.1-12.1	10-65	3.2-13.2		2.1-4.4.4	12-12.1				4-10.1		
11	80	74	80	13	66	13.3	all	80	46	80	68	12.12	11.1	1.2	7.1
		75-76	81-83	13.1-TP		13.4									

Notes Sub-features (1)

See full reference on MDN Web Docs

¹ Not supported on option element

Support data for this feature provided by: MDN browser-compat-data

CSS property: font-family: system-ui

Usage % of all users Global 68.6% + 20.75% = 89.35%

Current aligned Usage relative Date relative Apply filters Show all ?

IE	Edge	Firefox	Chrome	Safari	Opera	iOS Safari	Opera Mini	Android Browser	Opera Mobile	Chrome for Android
	12-18	2-42	4-55	3.1-8	10-42	3.2-8.4				
6-10	79	43-73	56-79	9-12.1	43-65	9-13.2		2.1-4.4.4	12-12.1	

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Questions?